

NGUYEN DINH PHONG

Bachelor of Science in Mechanical Engineering • Research Assistant • Lead of Project
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EDUCATION

Vin University, Hanoi,

Vietnam, 2023-2027

BSc. Mechanical Engineering

- Scholarships:

- + 95% tuition scholarship at VinUniversity Vietnam, 2023
- + Bronze Change Maker Scholarship; awarded for academic year 2023–2024 at VinUniversity Vietnam, 2024
- + Activator Scholarship Recipient; awarded visit to Bosch Global Software Technologies, (Spring) India, 2025
- + “Design Futures·Shared Vision” Scholarship; awarded research at Tsinghua University, (Summer) China, 2025

- Achievements:

- + **Top 1**, BOSCH Activator Scholarship 2024 by BOSCH Global Software Technologies Vietnam, 2025
- + **Global Nominee**, NASA International Space Apps Challenge 2025 (Vietnam) Vietnam, 2025
- + **Silver medal**, International Astronomy and Astrophysics Competition 2025 (IAAC) USA, 2025
- + **First Runner-Up**, Schneider Electric’s Future Impact Challenge 2025 Vietnam, 2025
- + **Runner-up**, Google Developer Groups On Campus – Hackathon Vietnam Vietnam, 2025
- + **2nd Runner-up**, RMIT Accessibility Design Competition Vietnam, 2024
- + **Top 10**, REIvolution Idea HKU Vietnam Innovation Contest by Hong Kong University Hong Kong, 2024
- + **Top 10**, Finalists of UAVS Startup Challenge 2024: CROSSNOVATION Australia, 2024
- + **Top 1**, DApp Track at Winter Ideathon by Web3 Space Vietnam, 2023
- + **Top 1**, Main Track at Solana Consumer Hack 11 by Solana Superteam Vietnam Vietnam, 2023
- + Multiple top prizes (**Top 1–Top 10**) in startup and innovation competitions Vietnam, 2023 - 2025

TECHNOLOGY RESEARCH EXPERIENCE

Thermal Management System for Battery Energy Storage System (BESS) *Center Environment Intelligence, Vietnam*
Research Assistant September 2025 - Present

- Built a multi-physics BESS simulation model, coupling electrochemical heat generation with Heat Transfer in Solids, Heat Transfer in Pipes, and Pipe Flow to capture conduction, convection, and Joule heating effects.
- Conducted time-dependent and stationary simulations at pack level to analyze temperature distribution, coolant flow, and pressure drop under varying charge/discharge conditions (up to 80% SOC).
- Modeled anisotropic prismatic cells, cooling plates, and liquid channels, enabling quantitative evaluation of thermal stability, cooling efficiency, and safety margins.

Multiphysics Simulation and Optimization of Resonant Megastructures – LEGO Project *Politecnico di Bari, Italy*
Research Intern January 2026 – March 2026

- Developed high-fidelity physics-based simulation models in COMSOL Multiphysics to analyze coupled field behavior and energy transport under geometric and material constraints.
Performed eigenfrequency and frequency-domain analyses, applying simulation-driven topology optimization via MATLAB–COMSOL LiveLink to improve system performance metrics.
Strengthened core skills in numerical modeling, solver setup, and multiphysics coupling, directly applicable to thermo-fluid, solid mechanics, and energy system simulations.

Lumos Project – Adaptive Interfaces for Martian Habitats

Tsinghua University, China

Research Collaborator, “Design Futures: Shared Vision” – Tsinghua University

Summer 2025

- Engineered bio-integrated habitat systems through parametric CAD modeling and form-finding optimization, targeting structural efficiency under extraterrestrial constraints.
- Performed multi-physics simulations (environmental load, thermal comfort, and material behavior) to validate enclosure adaptability and occupant well-being.
- Advanced a transdisciplinary design framework, integrating mechanical analysis with interactive systems and human-centered spatial strategies for scalable Martian habitats.

Nanophotonics Integration in Optoelectronics Lab (NIO Lab) - <https://niolab.net/#/>

Hanoi, Vietnam

Undergraduate Research Assistant

August 2025 - Present

- Designing ultrahigh-Q dielectric metasurfaces using topological optimization in COMSOL, automated via MATLAB LiveLink and assisted by LLM-based scripting.
- Integrating FEM-based electromagnetic modeling with inverse-design workflows to maximise Q-factor, near-field enhancement, and refractive-index sensitivity.
- Exploring AI-assisted photonic design automation using ChatGPT to streamline geometry generation, constraint handling, and documentation within metasurface simulation pipelines.

VinUni Biorobotics Lab - <https://bioroboticslaboratory.com/>

Hanoi, Vietnam

Undergraduate Research Assistant

October 2023 – June 2025

- Engineered a wearable haptic device that conveys directional cues via localized vibrotactile feedback across the back of the hand.
- Designed compact, modular mechanical architecture optimized for ergonomic fit, actuator integration, and extended wearability.
- Collaborated on multi-domain prototyping, focusing on the mechanical-electrical interface and human-centered feedback fidelity.

CommuniCare - Smart devices support the deaf and mute community

Hanoi, Vietnam

Leader, Lead of Hardware

January 2024 – September 2024

- Designed compact, ergonomic enclosures in SolidWorks for a handheld audio-to-text device supporting the hearing-impaired.
- Applied FEA to validate structural integrity under drop impact and daily use conditions.
- Integrated electronic components within mechanical constraints, ensuring usability, manufacturability, and thermal safety.

EcoFunds project

Hanoi, Vietnam

Hardware Researcher & IoT Developer

November 2023 – July 2024

- Engineered a solar-powered remote monitoring node integrating MPPT charge control, voltage & current sensing, and LiPo storage within an IP67 enclosure for off-grid deployment.
- Developed a LoRaWAN-based communication system (using Semtech SX126x + STM32 MCU) enabling low-power, long-range telemetry (>10 km), ideal for rugged highland environments.
- Designed custom PCBs and embedded firmware to interface sensors, implement power-efficient sleep cycles, and securely transmit data packets to a Lora WAN gateway, enhancing system autonomy and resilience.

INNOVATION AND LEADERSHIP EXPERIENCE

WIABLES - The project applies technology to products for people with disabilities

Hanoi, Vietnam

Founder and Lead of Organizer

June 2024 - Present

- Founded and led a global initiative uniting over 60 engineers and students in science and technology to co-develop assistive products using 3D printing, biomedical design, and user-centered engineering.
- Directed multi-team collaboration across Vietnam, India, and China, representing the project during an elite exchange at Bosch Global Software Technologies (India) and co-organizing a joint research program with Tsinghua University on digital innovation and AI entrepreneurship.
- Fostered partnerships with researchers and clinicians from institutions advancing applied research in health technology and accessible design for underserved populations.

WINDOU – Applying Technology Integrated Artificial Intelligence in Business of Recycled

Hanoi, Vietnam

Founder, Lead of Product

January 2024 – May 2025

- Initiated and led the development of an AI-powered platform for recyclable waste classification and home recycling service integration.
- Directed hardware-software prototyping and product research toward scalable, user-centered waste management solutions.
- Developed early-stage carbon credit linkage models through traceable recycling contributions.

Vietnam Association of Medical Education (VAME), 2024

Hanoi, Vietnam

Head of Academic Subcommittee Support

October 2024 – March 2025

- Led academic coordination for Vietnam's largest medical forum with 1,000+ participants, managing submissions and researcher engagement.

- Supported data compilation, translation, and logistics for presenters across disciplines.
- Collaborated with medical professionals to advance innovation in surgical systems and patient-centered care.

SPECIALIZED SKILLS

- **3D modeling and prototyping:** SolidWorks, Fusion 360, mechanical design for manufacturability, 3D printing (FDM/SLA).
- **Simulation and engineering analysis:** ANSYS, COMSOL Multiphysics, FEA, thermal and structural analysis...
- **Mathematical modeling and research:** MATLAB, Simulink, LaTeX, experimental design, data processing.
- **Technical communication and interdisciplinary teamwork:** specialized English for engineering, academic writing, pitching, and presentation.
- **Language:** Vietnamese (Native), English (Advanced)

SUPPORTING DOCUMENTS

- **Certificates & Honors:** [Link 1](#)
- **Projects & Portfolios:** [Link 2](#)
- **LinkedIn:** [Link 3](#)